

Thin-lens equation

$$\frac{1}{s_o} + \frac{1}{s_i} = (n_l - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

The image focal length f_i

$$\lim_{s_o \rightarrow \infty} s_i = f_i = \frac{R_2 R_1}{(n_l - 1)(R_2 - R_1)}$$

The object focal length f_o

$$\lim_{s_i \rightarrow \infty} s_o = f_o = \frac{R_2 R_1}{(n_l - 1)(R_2 - R_1)}$$

Gaussian lens formula

$$\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f}$$

where

$$\frac{1}{f} = (n_l - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Example: compute the focal length in air of a thin planar-convex lens having a radius of curvature of 50 mm and an index of 1.5.

On the planar surface: $R_1 = \infty$, $R_2 = -50$

$$\frac{1}{f} = (1.5 - 1) \left(\frac{1}{\infty} - \frac{1}{-50} \right), f = 100 \text{ mm}$$

On the curved surface: $R_1 = 50$, $R_2 = \infty$

$$\frac{1}{f} = (1.5 - 1) \left(\frac{1}{50} - \frac{1}{\infty} \right), f = 100 \text{ mm}$$

If an object is alternately placed at distances 600 mm, 200 mm, 150mm, 100mm, and 50 mm from the lens on either side, where can we find the image points.

$$s_i = \frac{s_o f}{s_o - f} = \frac{100 s_o}{s_o - 100}$$

$$s_i = 120, 200, 300, \infty, \text{ and } -100 \text{ mm}$$

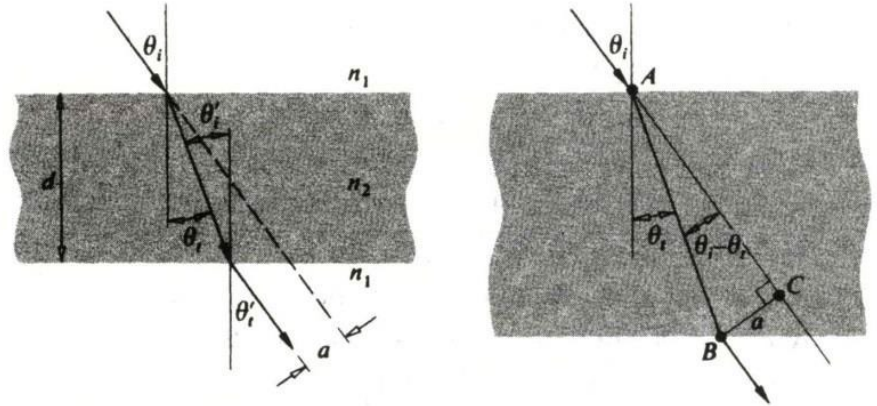
Optical Center

A beam entering a planar transparent plate emerges parallel to its initial direction

$$\theta_i = \theta_t$$

The lateral displacement of the beam

$$a = \frac{d \sin(\theta_i - \theta_t)}{\cos \theta_t}$$



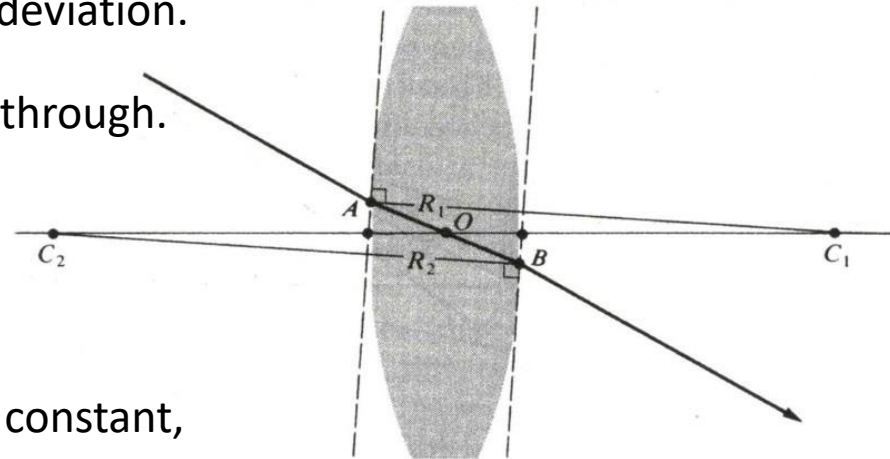
A ray traversing a medium bounded by parallel planes will be displaced laterally but will suffer no angular deviation.

Optical center: the point that all such rays pass through.

Triangles AOC_1 and BOC_2 are similar

$$|R_1|(\overline{OC_2}) = |R_2|(\overline{OC_1})$$

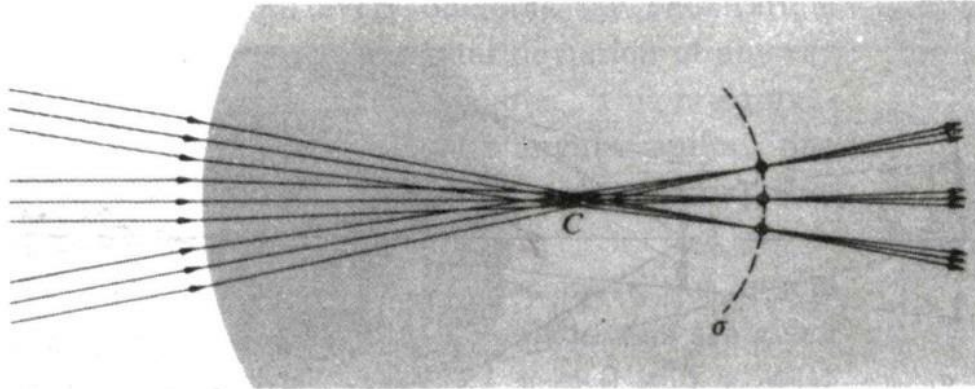
Since the radii are constant, the location of O is constant, independent of A and B.



For thin lens, rays passing through O may be drawn as straight lines.

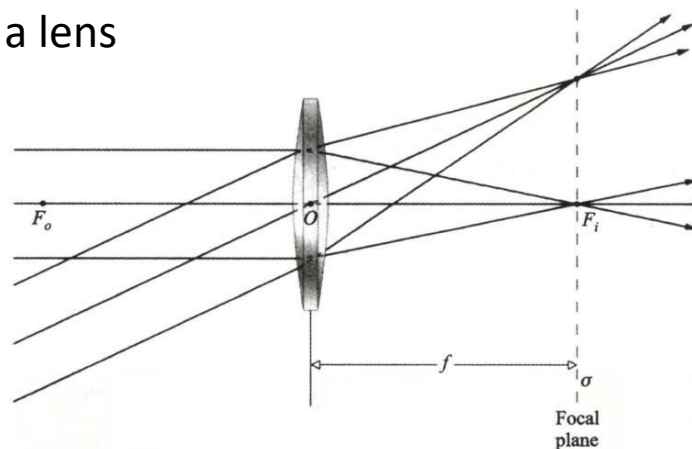
Focal Points and Planes

Focal plane of a spherical refracting surface

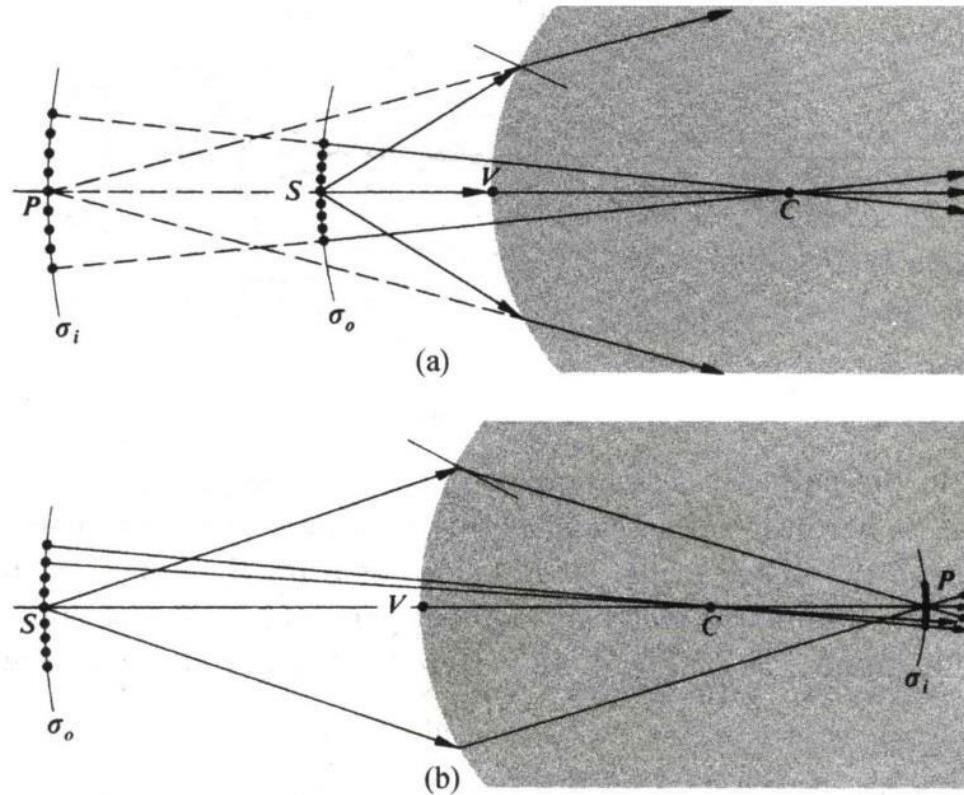


Since the ray cone must indeed be narrow, σ can be represented as a plane normal to the symmetry axis and passing through the image focus.

Similarly, focal plane of a lens



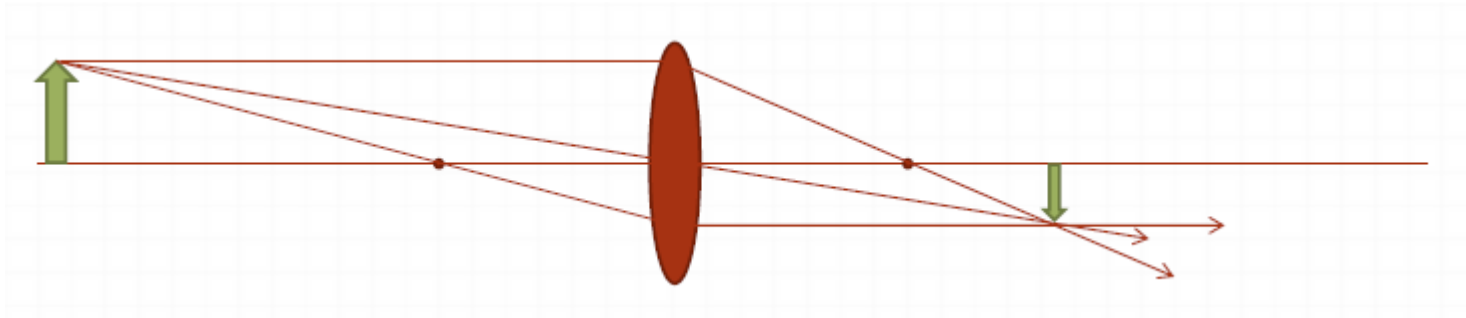
Finite imagery



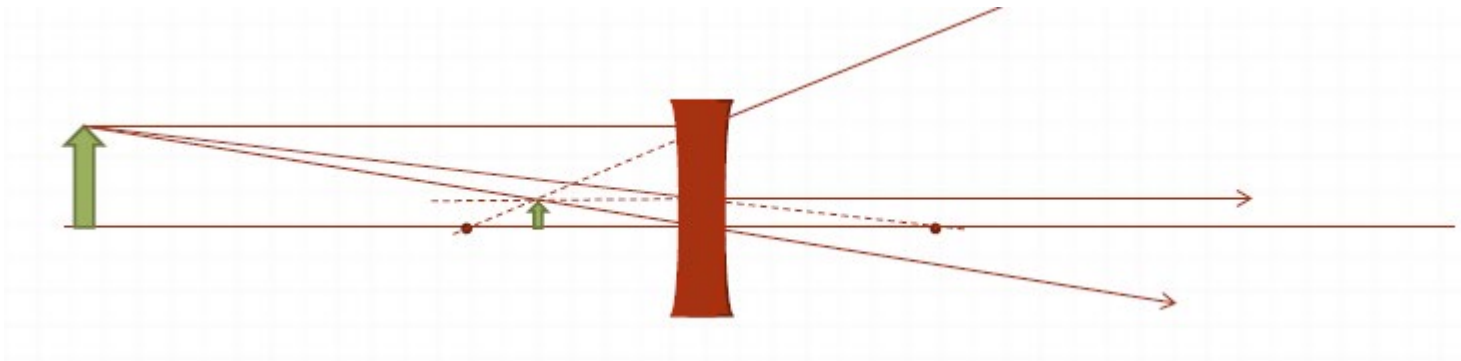
The final image formed by a lens of a small planar object normal to the optical axis will itself be a small plane normal to that axis

Ray diagrams

A real object and a positive lens



A real object and a negative lens



Triangles AOF_i and $P_2P_1F_i$ are similar.

$$\frac{y_o}{|y_i|} = \frac{f}{s_i - f}$$

Triangles S_2S_1O and P_2P_1O are similar.

$$\frac{y_o}{|y_i|} = \frac{s_o}{s_i}$$

So $\frac{s_o}{s_i} = \frac{f}{s_i - f}$ **Gaussian lens formula**

$$\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f}$$

Triangles $S_2S_1F_o$ and BOF_o are similar.

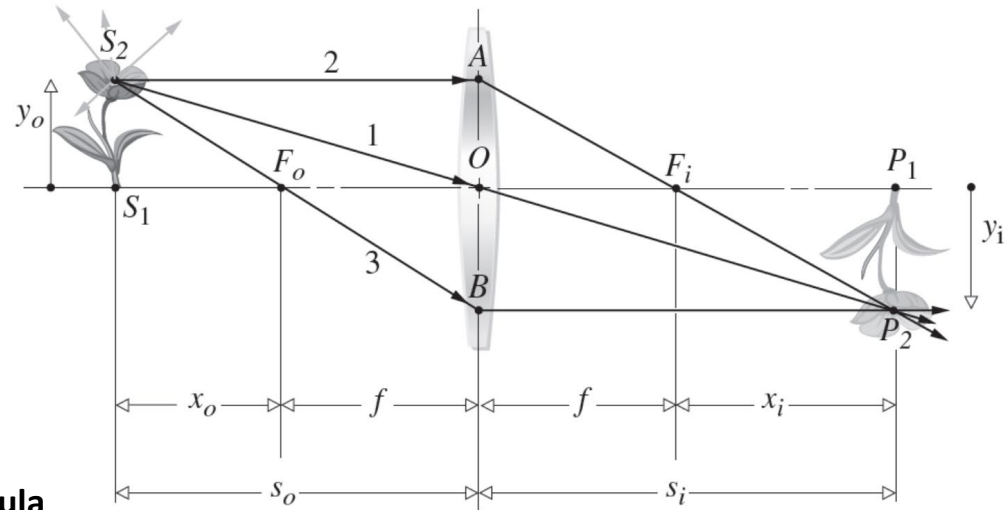
$$\frac{f}{s_o - f} = \frac{|y_i|}{y_o}$$

$$s_o - f = x_o \quad s_i - f = x_i$$

Newtonian form of the lens equation

$$x_o x_i = f^2$$

The object and image must be on opposite sides of their respective focal points



Sign convention for spherical refracting surfaces and thin lenses
(light entering from the left)

s_o, f_o : + left of V

x_o : + left of F_o

s_i, f_i : + right of V

x_i : + right of F_i

R : + if C is right of V

y_o, y_i : + above optical axis

Transverse magnification $M_T = \frac{y_i}{y_o}$

A positive M_T connotes an erect image, while a negative value means the image is inverted.

s_i and s_o are both positive for real objects and images, then all real images formed by a single thin lens will be inverted.

$$M_T = -\frac{s_i}{s_o} = -\frac{x_i}{f} = -\frac{f}{x_o}$$

Thin-lens equation $\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f}$

As the object approaches the lens, the real image moves away from it.

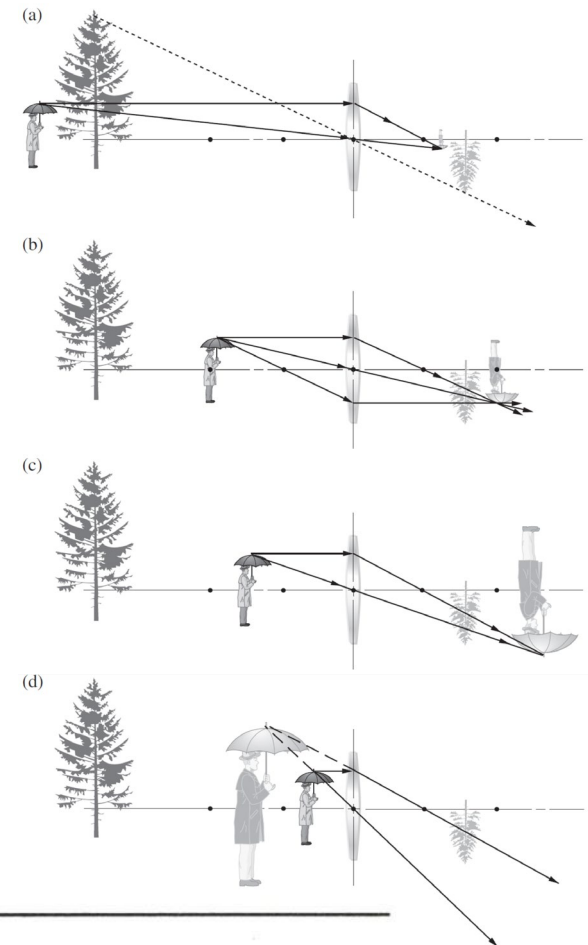
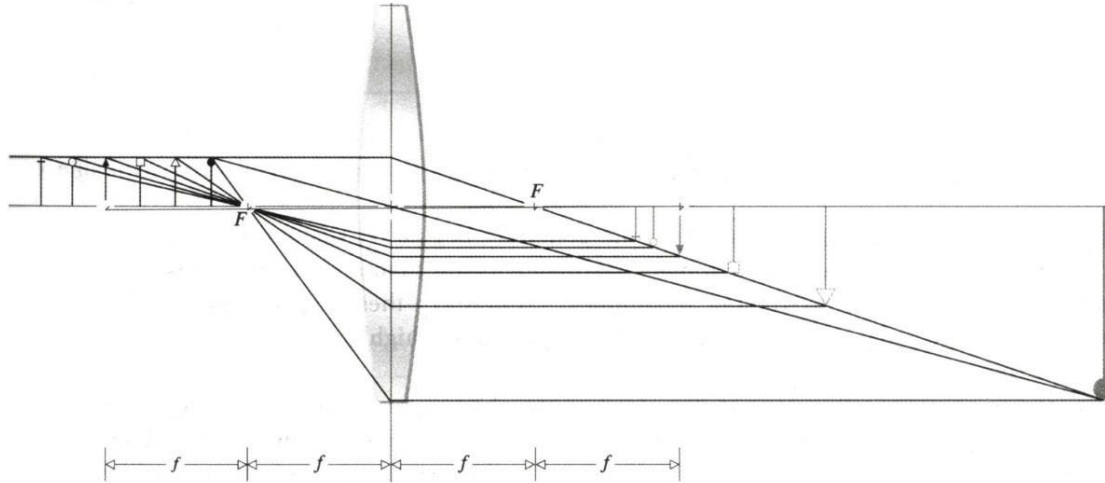


TABLE 4.3 Images of Real Objects Formed by Thin Lenses

Convex				
Object	Image			
Location	Type	Location	Orientation	Relative Size
$\infty > s_o > 2f$	Real	$f < s_i < 2f$	Inverted	Minified
$s_o = 2f$	Real	$s_i = 2f$	Inverted	Same size
$f < s_o < 2f$	Real	$\infty > s_i > 2f$	Inverted	Magnified
$s_o = f$		$\pm \infty$		
$s_o < f$	Virtual	$ s_i > s_o$	Erect	Magnified

The transformation from object to image space is not linear:

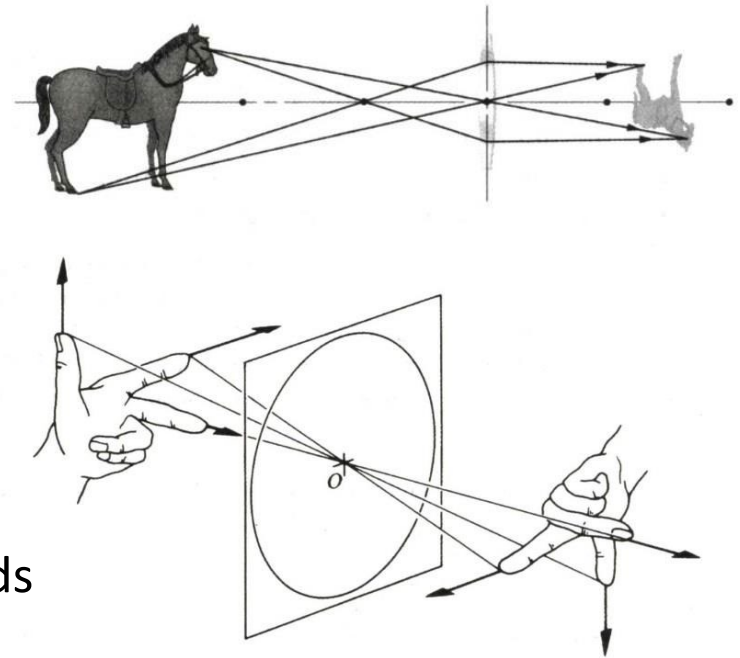


Longitudinal magnification

$$M_L \equiv \frac{dx_i}{dx_o} = -\frac{f^2}{x_o^2} = -M_T^2$$

By differentiating $x_o x_i = f^2$

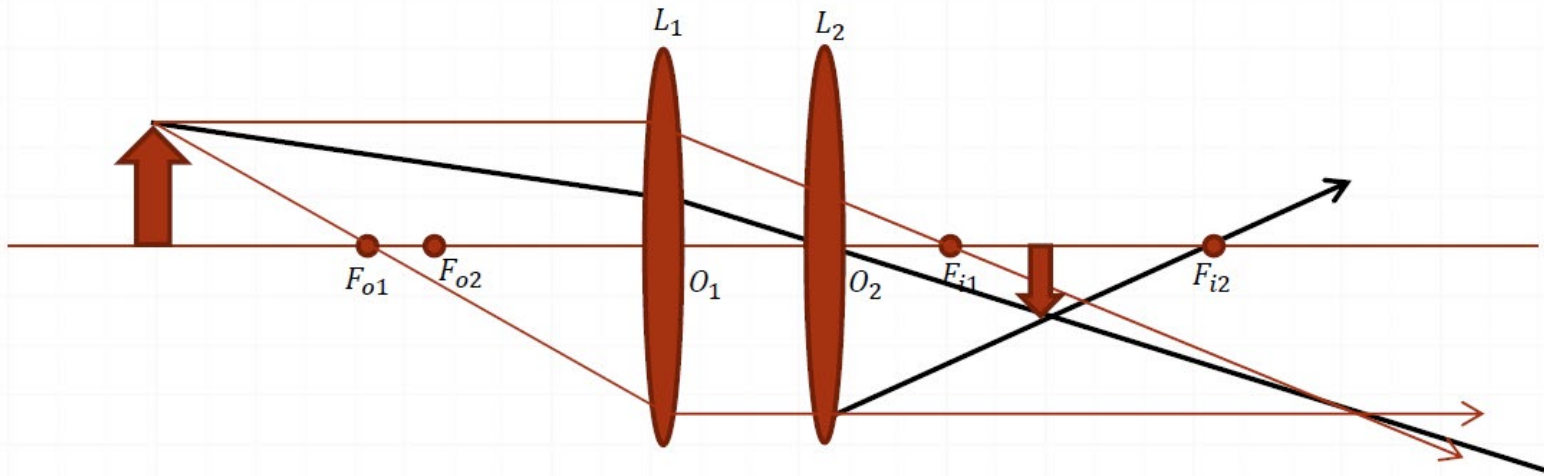
Evidently, $M_L < 0$: a positive dx_o corresponds to a negative dx_i and vice versa.



Thin-Lens Combinations

Consider two thin positive lenses L_1 and L_2 separated by a distance d , which is smaller than either focal length.

The resulting image can be located graphically as follows

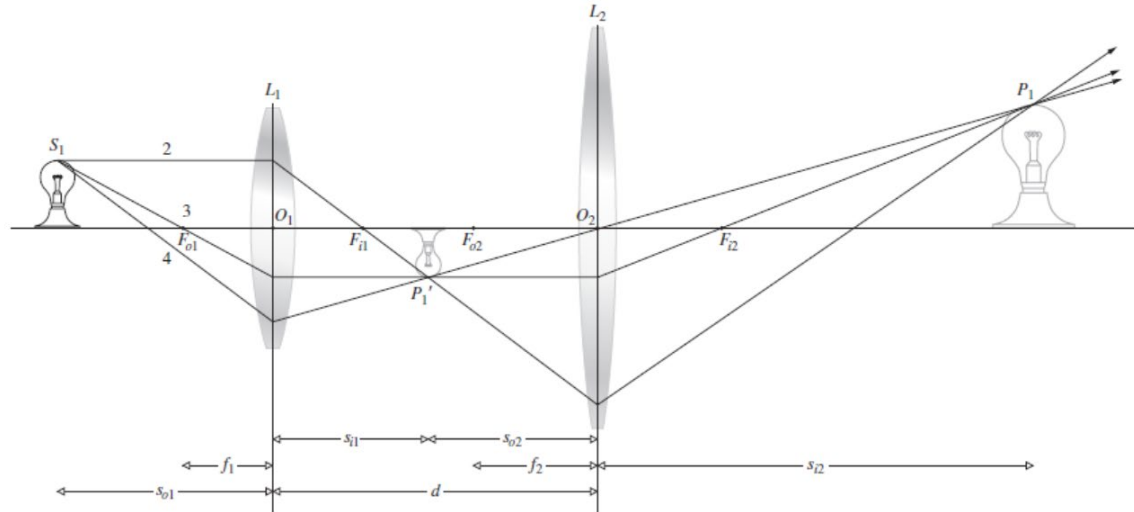


Consider a similar pair of lenses (increased separation)

$$\text{For } L_1 \quad \frac{1}{s_{i1}} = \frac{1}{f_1} - \frac{1}{s_{o1}}$$

$$\text{For } L_2 \quad \frac{1}{s_{i2}} = \frac{1}{f_2} - \frac{1}{s_{o2}}$$

$$s_{o2} = d - s_{i1}$$



If $d > s_{i1}$, the object for L_2 is real.

If $d < s_{i1}$, the object for L_2 is virtual.

$$s_{i2} = \frac{f_2 d - f_2 s_{o1} f_1 / (s_{o1} - f_1)}{d - f_2 - s_{o1} f_1 / (s_{o1} - f_1)}$$

$$M_T = M_{T1} M_{T2} = \frac{f_1 s_{i2}}{d(s_{o1} - f_1) - s_{o1} f_1}$$

Example: compute the image distance associated with an object place 50 cm from the first of two positive lenses. These in turn are separated by 20 cm and have focal lengths of 30 cm and 50 cm, respectively.

$$s_{i2} = \frac{50(20) - 50(50)(30)/(50 - 30)}{20 - 50 - 50(30)/(50 - 30)} = 26.2 \text{ cm}$$

Front focal length

If we set $s_{i2} \rightarrow \infty$, s_{o2} approaches f_2

So $s_{i1} \rightarrow d - f_2$

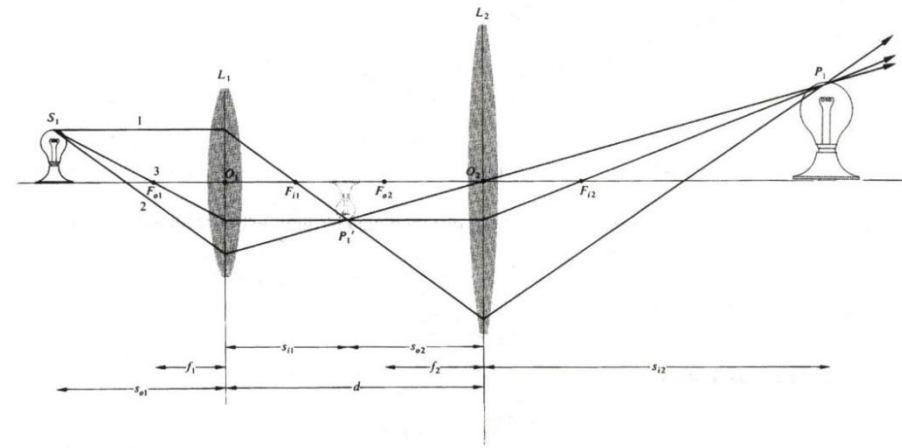
$$\left. \frac{1}{s_{o1}} \right|_{s_{i2} \rightarrow \infty} = \frac{1}{f_1} - \frac{1}{(d - f_2)} = \frac{d - (f_1 + f_2)}{f_1(d - f_2)}$$

$$\text{f. f. l.} = \frac{f_1(d - f_2)}{d - (f_1 + f_2)}$$

Back focal length

If we set $s_{o1} \rightarrow \infty$, $(s_{o1} - f_1) \rightarrow s_{o1}$

$$s_{i2} = \frac{f_2 d - f_2 s_{o1} f_1 / (s_{o1} - f_1)}{d - f_2 - s_{o1} f_1 / (s_{o1} - f_1)} = \frac{f_2(d - f_1)}{d - (f_1 + f_2)} = \text{b. f. l.}$$



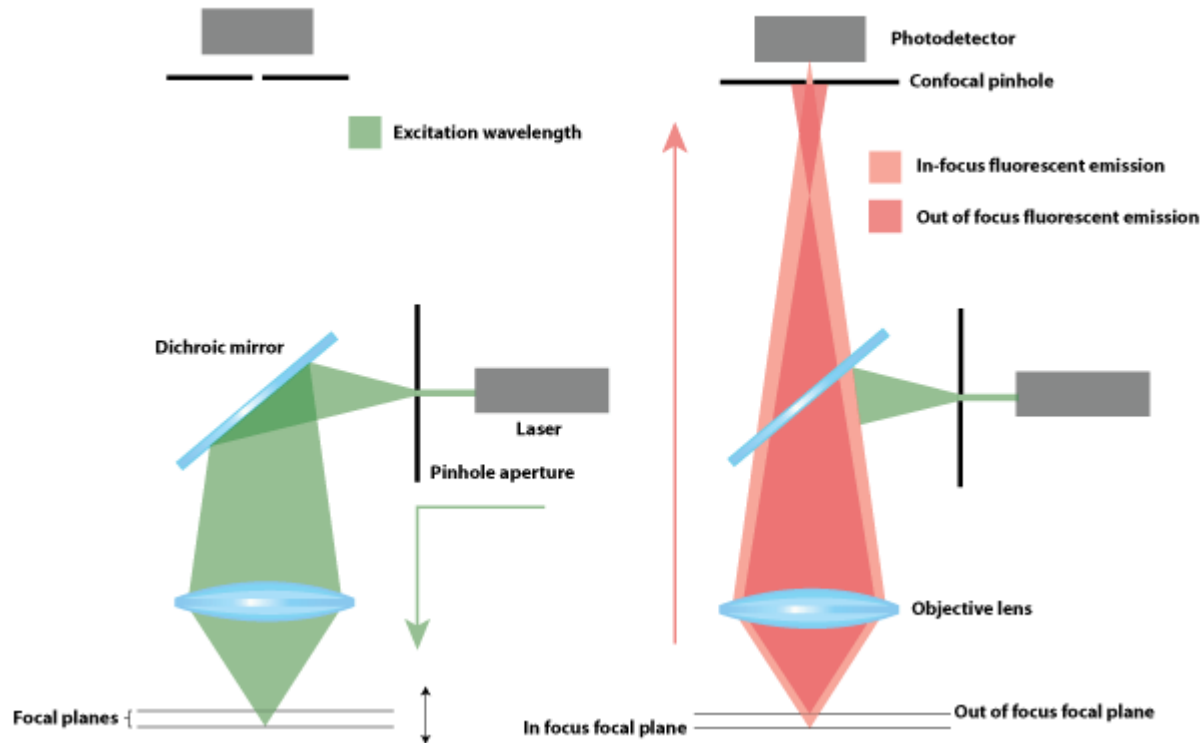
For $d \rightarrow 0$

$$\text{f. f. l.} = \text{b. f. l.} = \frac{f_1 f_2}{f_1 + f_2}$$

Effective focal length

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

Confocal microscopy



- ❑ The confocal microscope is so called because the illumination pinhole, the plane of focus within the specimen and the detector pinhole are all situated at conjugate focal planes. By restricting the illumination to a pinhole, rather than illuminating the entire field of view at once it becomes possible to build up a blur-free image in three dimensions.

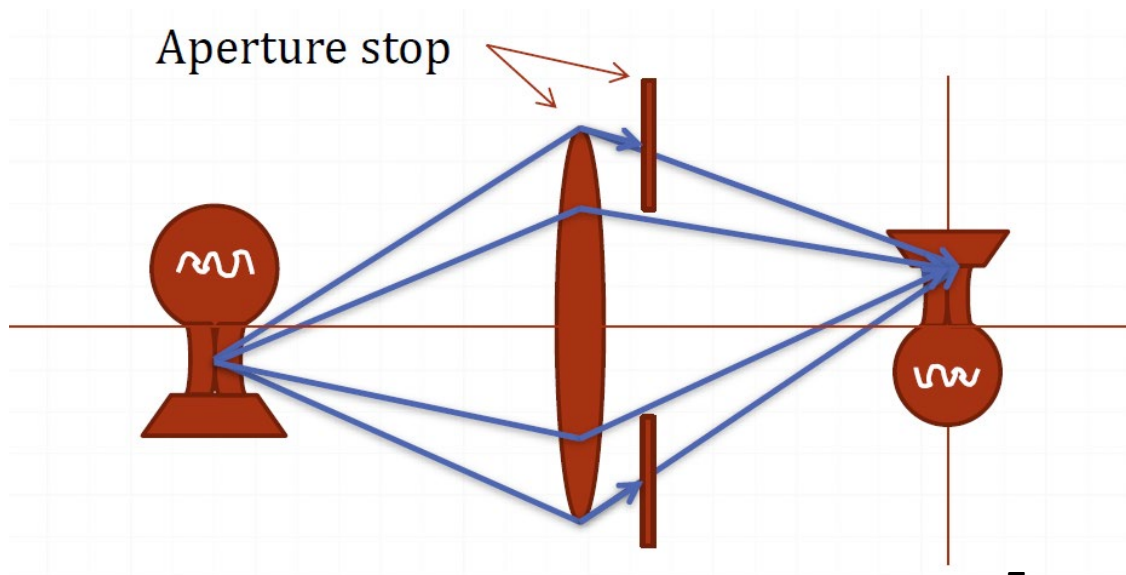
Homework

Problems: 5.8, 5.13, 5.15, 5.24, 5.35, 5.37

4.3 Stops (光阑)

Aperture and Field Stops

The intrinsically finite nature of all lenses demands that they collect only a fraction of the energy emitted by a point source.



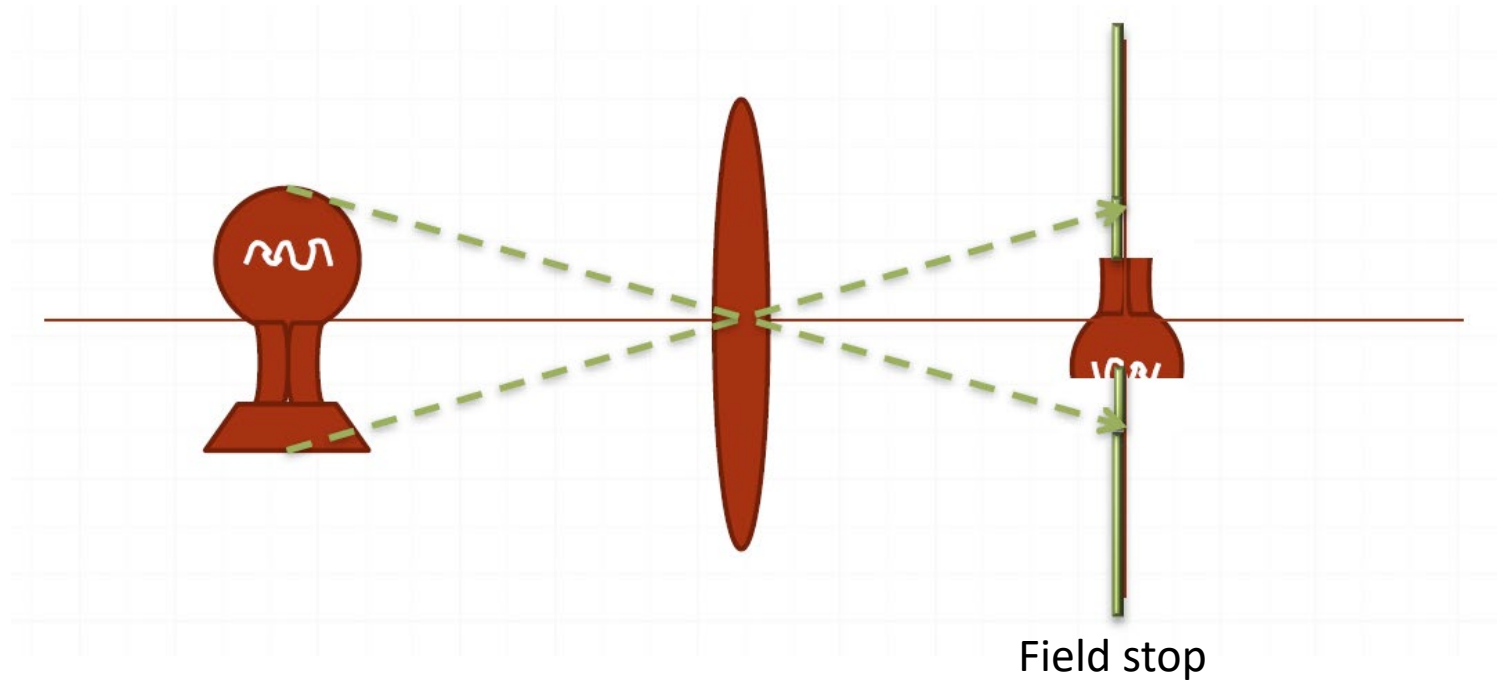
Aperture stop (A.S., 孔径光阑): any element, be it the rim of a lens or a separate diaphragm, that determines *the amount of light reaching the image*.

Example:
diameter of the lens



The adjustable
leaf diaphragm





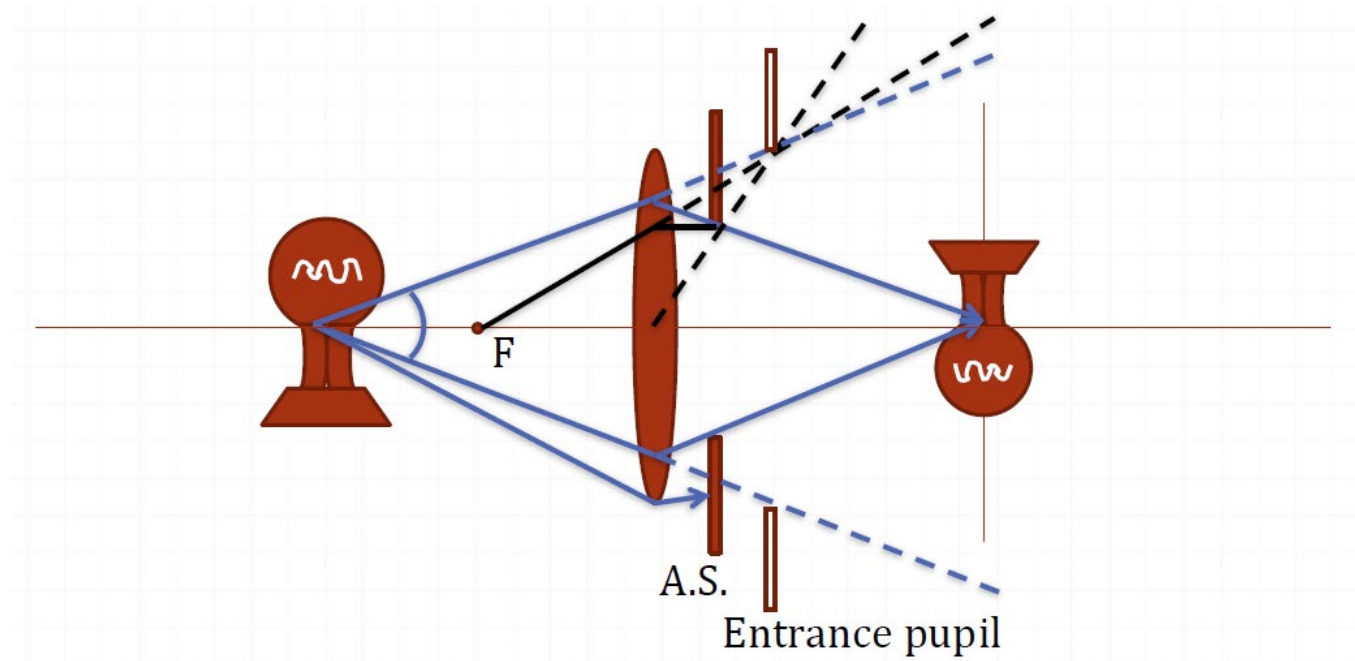
Field stop (F. S., 视场光阑): the element limiting the size or angular breadth of the object that can be imaged by the system. It determines the field of view of the instrument.

Example: the edge of CCD or film



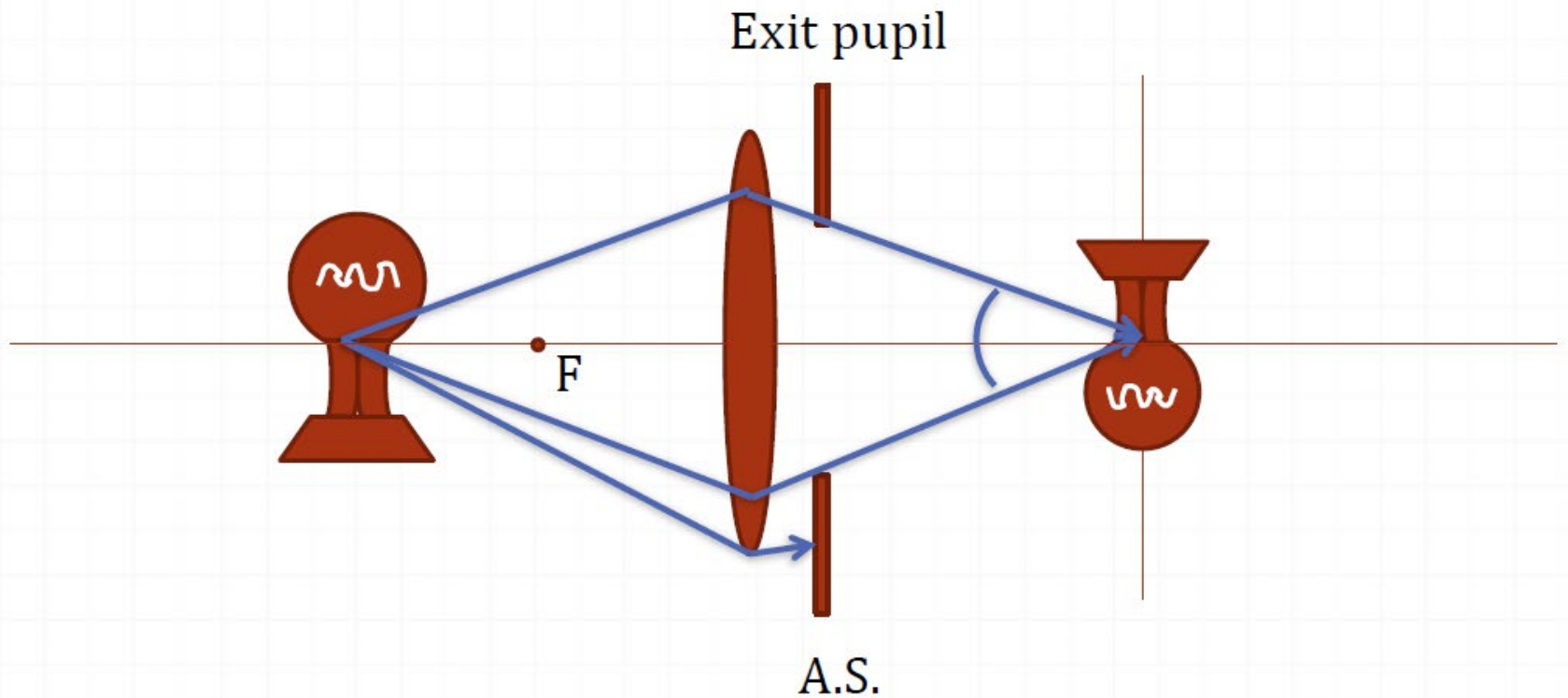
Pupils (光瞳)

Pupil: determine whether or not a given ray will traverse the entire optical system, namely an image of the aperture stop.



Entrance pupil of a system: the image of the aperture stop as seen from an axial point on the object through those elements preceding the stop.

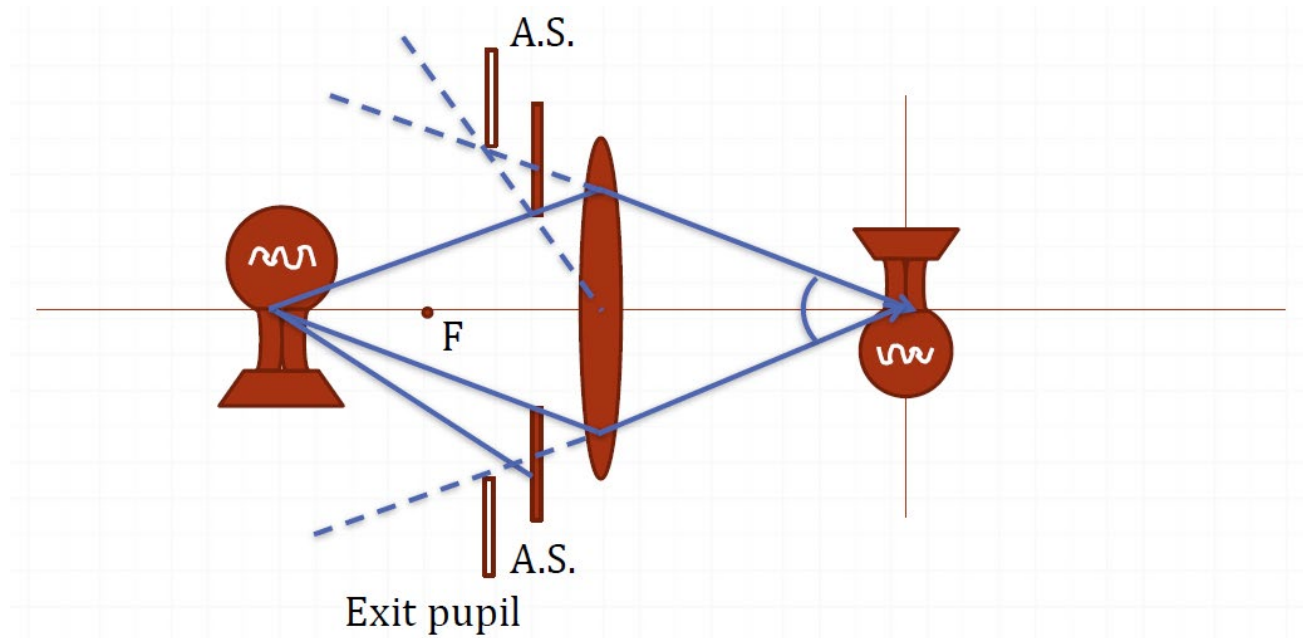
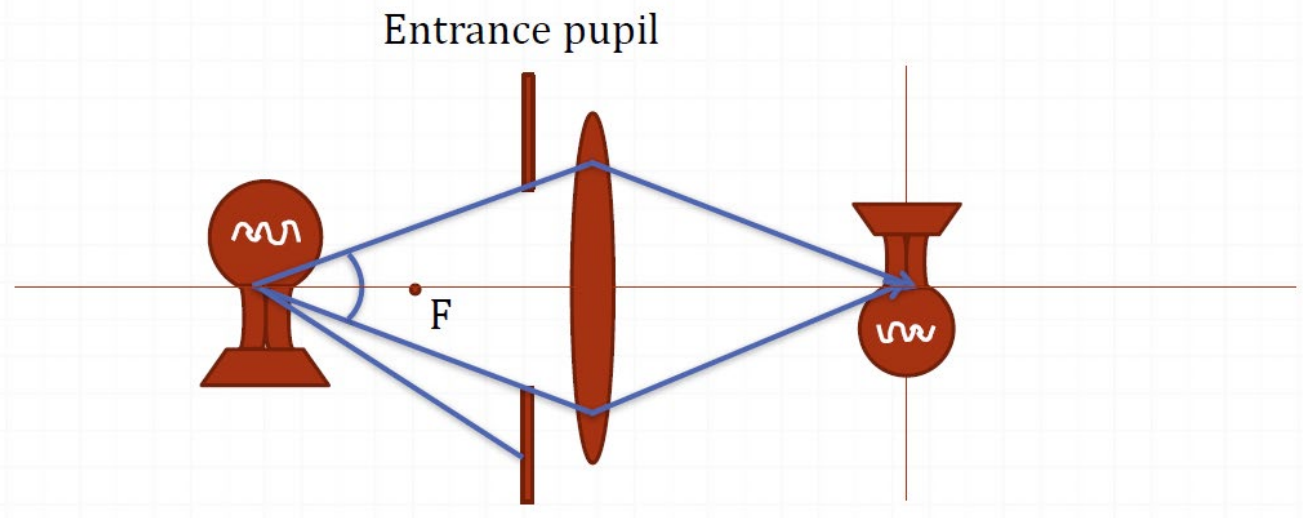
Function: the cone of light actually entering the optical system is determined by the entrance pupil.



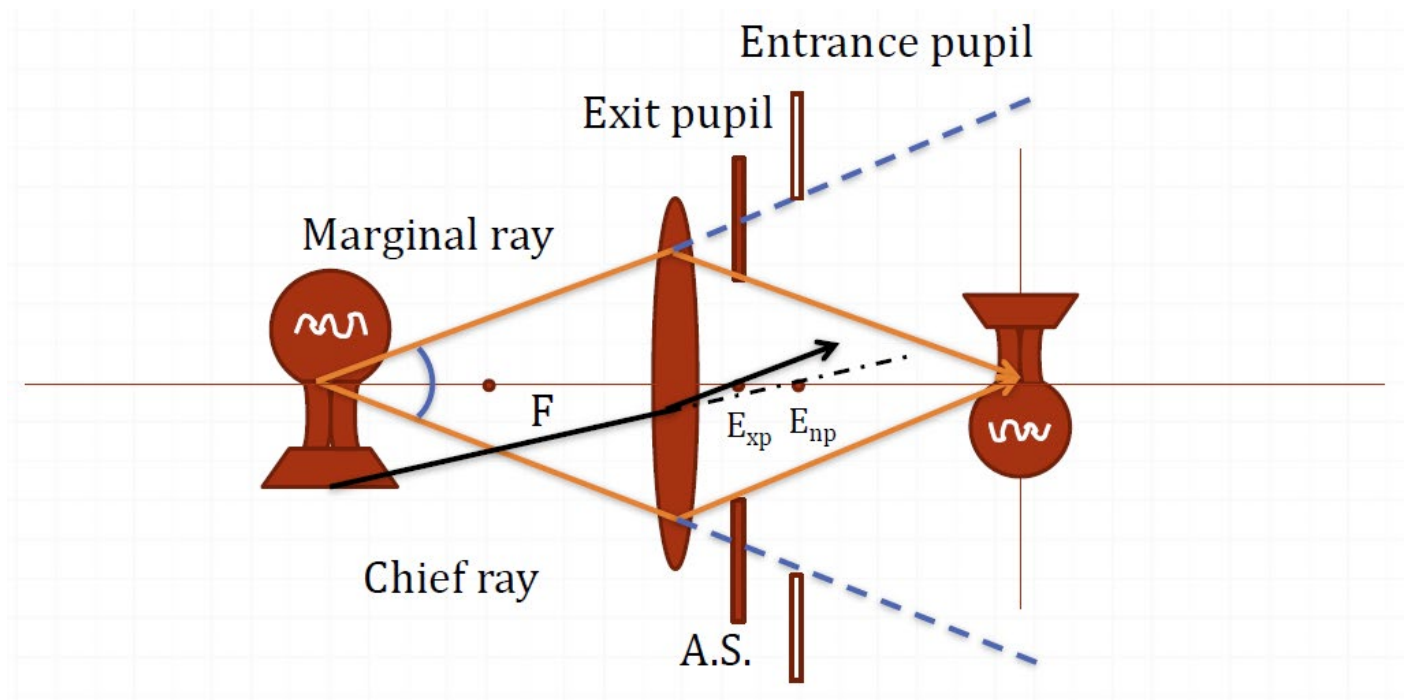
Exit pupil: the image of the A. S. as seen from an axial point on the image plane through the interposed lenses, if there are any.

Function: the cone leaving it is controlled by the exit pupil.

Example: a front aperture stop F



Chief and Marginal Ray



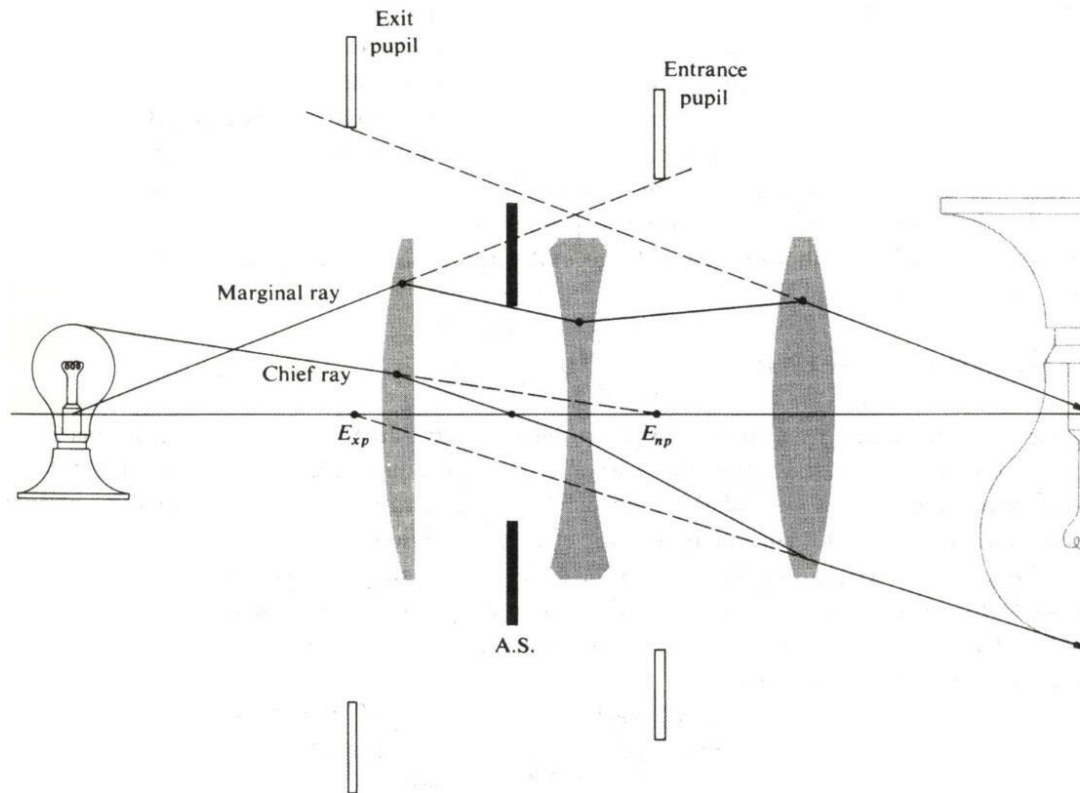
Chief ray: any ray from an off-axis object point that passes through the center of the aperture stop.

The chief ray enters the optical system along a line directed toward the midpoint of the entrance pupil, E_{np} . And leaves the system along a line passing through the center of the exit pupil, E_{xp} .

Marginal ray: it goes from the axial object point to the rim or margin of the entrance pupil.

Entrance pupil for optical system

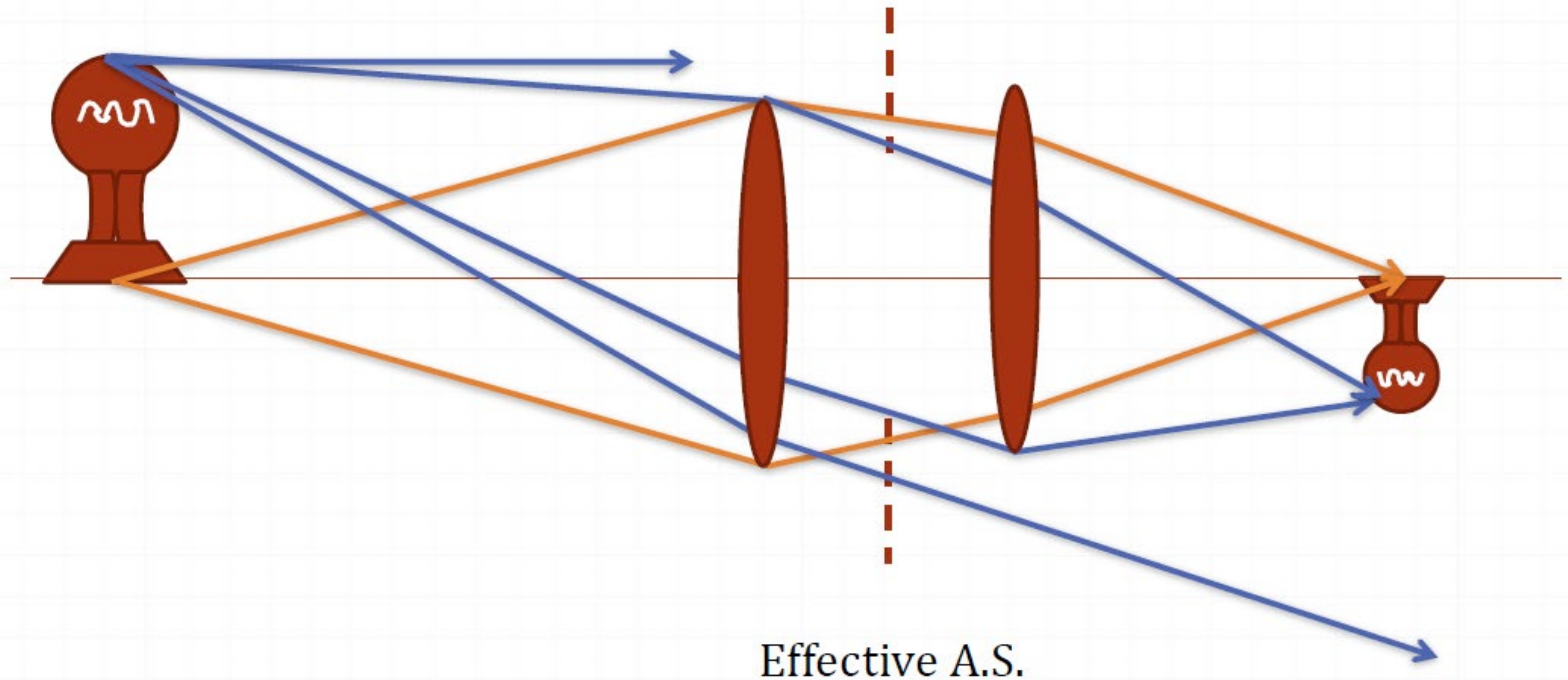
For a complex system, each component of the system must be imaged by the remaining elements to its left. The image that subtends the smallest angle at the axial object point is the **entrance pupil**.



Pupils and stops for a three-lens system

Vignetting (渐晕效应)

The cone of rays that can reach the image plane becomes narrower due to the effective aperture stop, then the image is gradually faded out at points near its periphery.



Relative Aperture (相对孔径)

Irradiance or flux density (the energy per unit area per unit time)

Proportional to **area of the entrance pupil**

Inversely proportional to **image area**

The entrance pupil area $\propto$ square of its diameter (D^2)

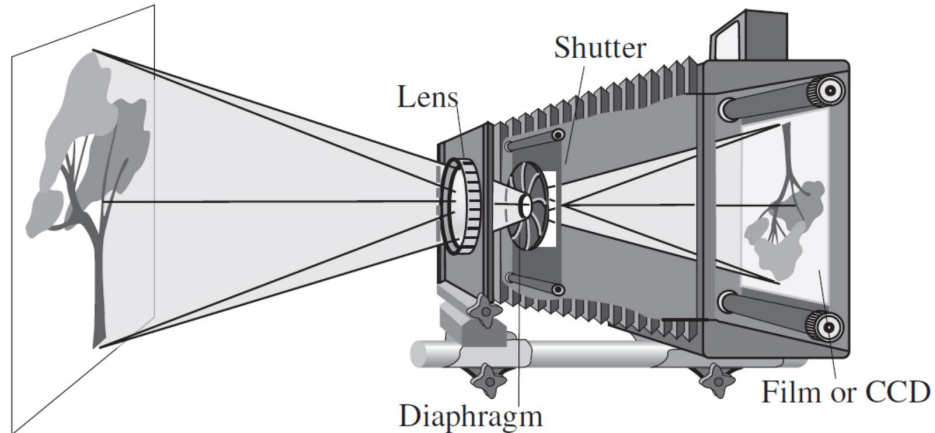
$$\text{Since } M_T \equiv \frac{|y_i|}{y_o} = -\frac{x_i}{f} = -\frac{f}{x_o}$$

The image area $\propto$ square of focus (f^2)

Irradiance or flux density at the image plane

$$I \propto \left(\frac{D}{f}\right)^2$$

where D/f is known as the **relative aperture**



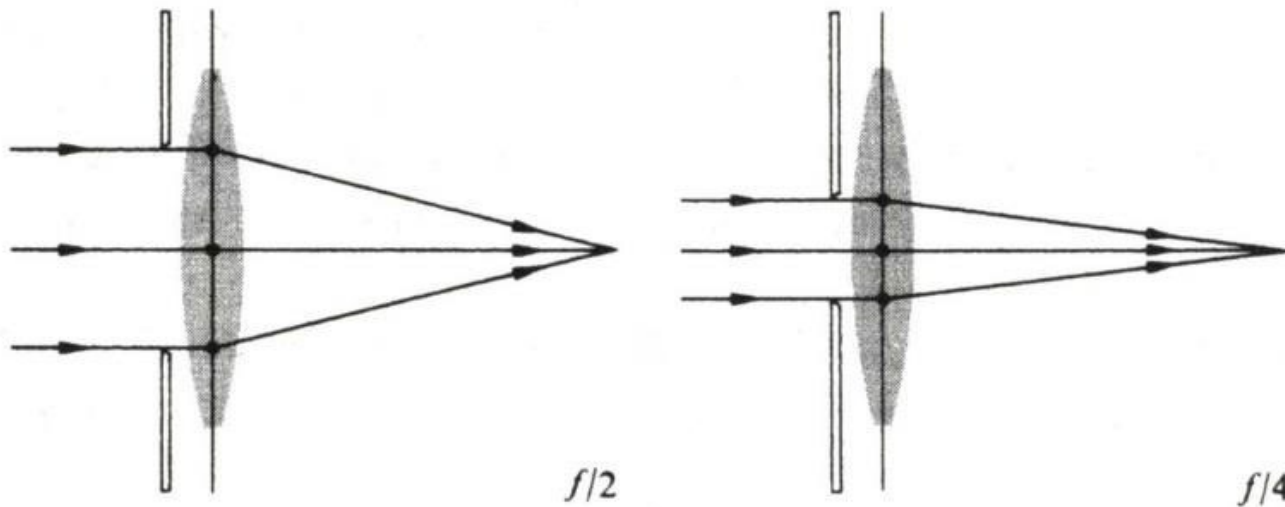
***f*-Number**

Focal ratio or f-number ($f/\#$): inverse of relative aperture

$$f/\# = \frac{f}{D}$$

Example: a lens with a 25 mm aperture and a 50 mm focal length

$$f/\# = \frac{50}{25} = 2$$



A smaller f-number clearly permits more light to reach the image plane.

Application

Camera lenses are usually specified by their focal lengths and the largest possible apertures



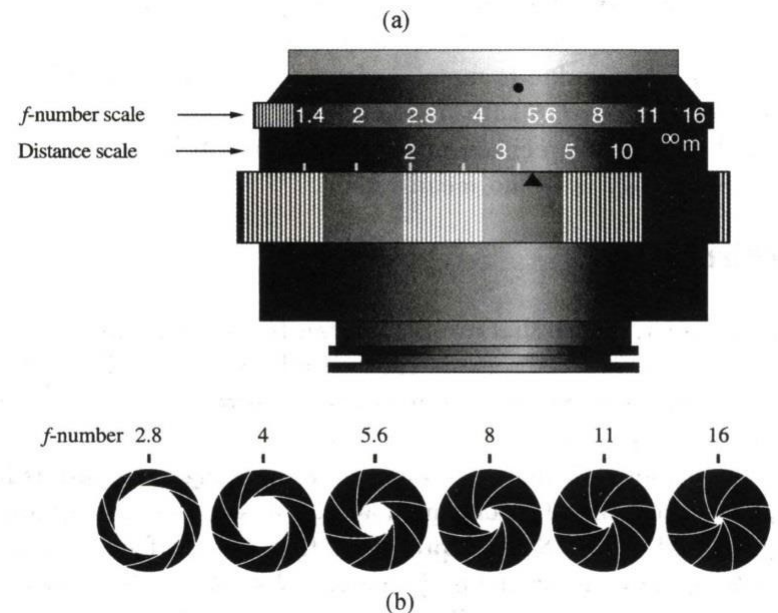
$$\text{Irradiance} \propto \left(\frac{D}{f}\right)^2 = \text{square of relative aperture}$$

$$\text{Exposure time} \propto \left(\frac{f}{D}\right)^2 = \text{square of } f/\#$$

The speed of the lens

Lens diaphragms have f-number markings of 1, 1.4, 2, 2.8, 4, 5.6, 8, 11, 16, 22, and so on.

The same amount of light will reach the film whether the camera is set for f/1.4 at 1/500th of a second, f/2 at 1/250th of a second, or f/2.8 at 1/125th of a second.



4.4 Mirrors

A mirror: a piece of black glass or a finely polished metal surface



Au mirror



Al mirror

For avoiding the losses due to metal surfaces, mirrors formed of multilayered dielectric films are used.

Broadband multilayer dielectric mirrors at 485-695 nm



Planar Mirrors

Law of reflection $\theta_i = \theta_r$

$$\theta_i = \theta_r$$

The image P and object S are equidistant from the surface

$$|s_o| = |s_i|$$

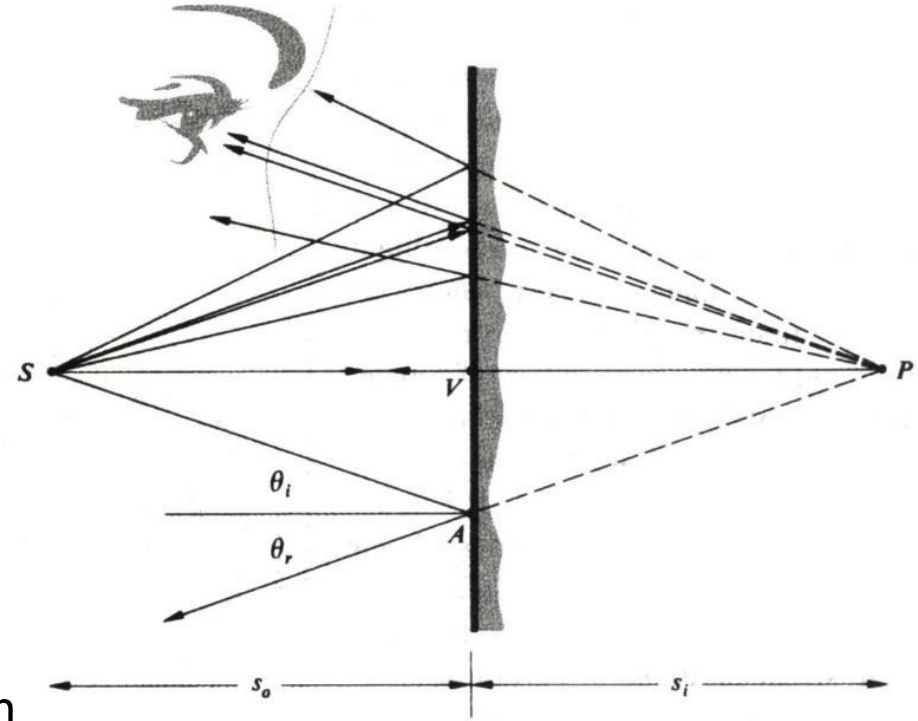
Follow the idea of virtual object and image distances being negative.

Definition: s_o and s_i are negative when they lie to the right of the vertex V .

Gaussian lens equation

$$\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f}$$

A life-size, virtual, erect image

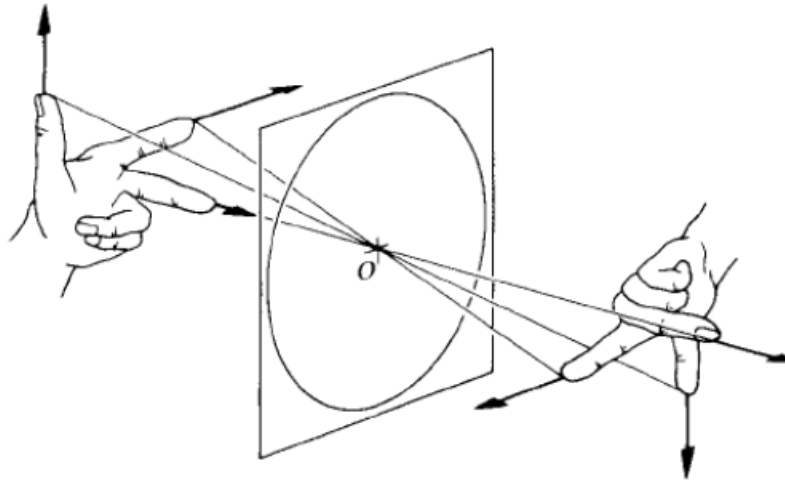


Transverse magnification

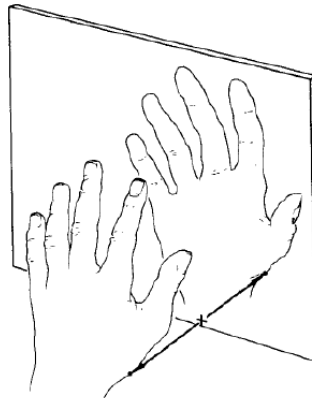
$$M_T = -\frac{s_i}{s_o} = +1$$

Reversion (反转) and Inversion (反演)

Reversion: 180° rotation about the optical axis

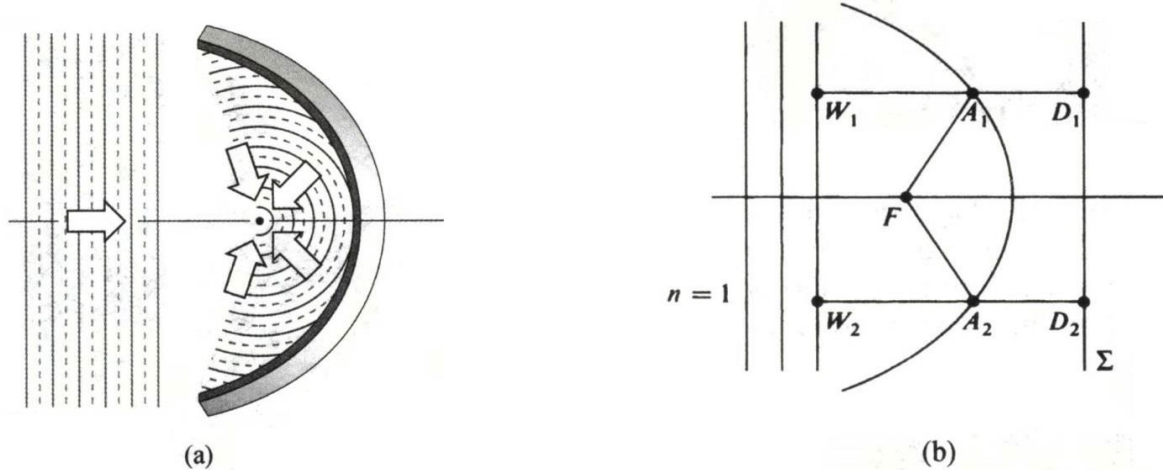


Inversion: convert a right-handed coordinate system in the object space into a left-handed one in the image space.



Aspherical Mirrors

Let's determine the configuration a mirror must have if an incident plane wave is to be reformed upon reflection into a converging spherical wave.



The optical path lengths for all rays must be equal:

$$OPL = \overline{W_1 A_1} + \overline{A_1 F} = \overline{W_2 A_2} + \overline{A_2 F}$$

A plane Σ parallel to the incident wavefronts

$$\overline{W_1 A_1} + \overline{A_1 D_1} = \overline{W_2 A_2} + \overline{A_2 D_2}$$

There will be a surface for which $\overline{A_1 F} = \overline{A_1 D_1}$ and $\overline{A_2 F} = \overline{A_2 D_2}$, it is a **parabola**.
(抛物线)

Application

Automobile headlight reflectors



Microwave horns



Communications antennas



Spherical Mirrors

Equation for circular cross section of a sphere

$$y^2 + (x - R)^2 = R^2$$

Solution (concern the value x less than R)

$$x = R - (R^2 - y^2)^{1/2}$$

Expansion in binomial series

$$x = \frac{y^2}{2R} + \frac{1y^4}{2^2 2! R^3} + \frac{1 \cdot 3y^6}{2^3 3! R^5} + \dots$$

The first term: parabola

Other terms: deviation

For parabola

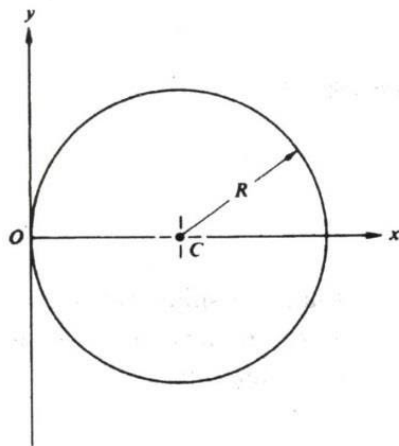
$$y^2 = 4fx$$

In paraxial region

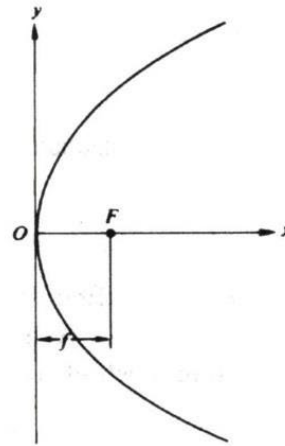
$$x = \frac{y^2}{2R}$$

where $f = R/2$

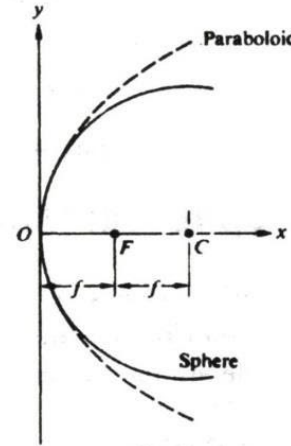
In the paraxial region, that is, in the immediate vicinity of the central axis, these two configurations will be essentially indistinguishable.



(a)



(b)



(c)

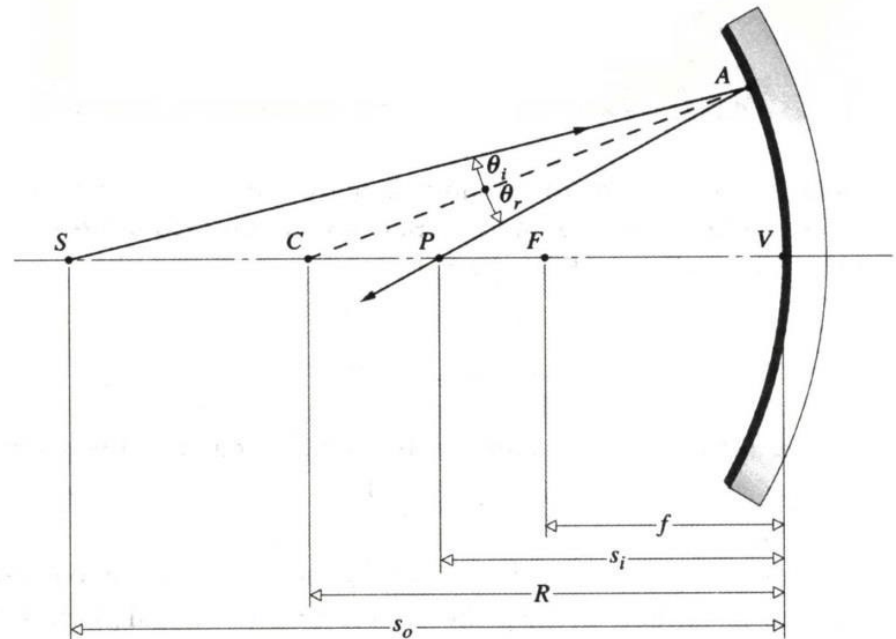
The mirror formula

Since $\theta_i = \theta_r$ $\frac{\overline{SC}}{\overline{SA}} = \frac{\overline{CP}}{\overline{PA}}$

Consider the sign of each item

$$\overline{SC} = s_o - |R| = s_o + R$$

$$\overline{CP} = |R| - s_i = -(s_i + R)$$



In the paraxial region

$$\overline{SA} \approx s_o$$

$$\overline{PA} \approx s_i$$

We have

$$\frac{1}{s_o} + \frac{1}{s_i} = -\frac{2}{R}$$

Sign Convention for Spherical Mirrors

Quantity

Sign

+

-

s_o

Left of V, real object

Right of V, virtual object

s_i

Left of V, real image

Right of V, virtual image

f

Concave mirror

Convex mirror

R

C right of V, convex

C left of V, concave

y_o

Above axis, erect object

Below axis, inverted object

y_i

Above axis, erect image

Below axis, inverted image

Focus

$$\frac{1}{s_o} + \frac{1}{s_i} = -\frac{2}{R}$$

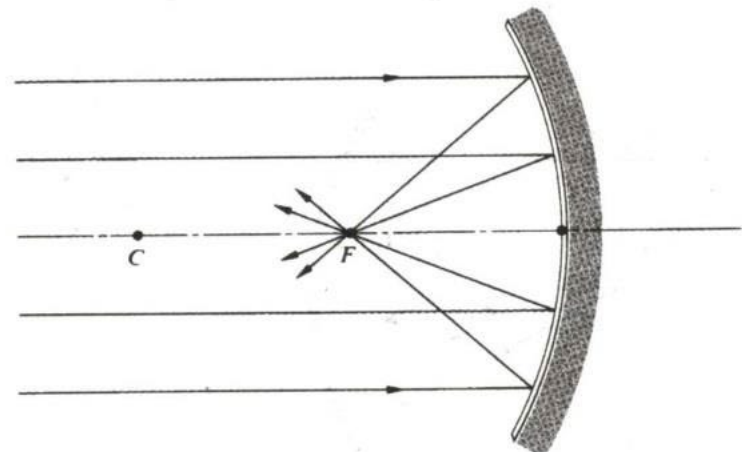
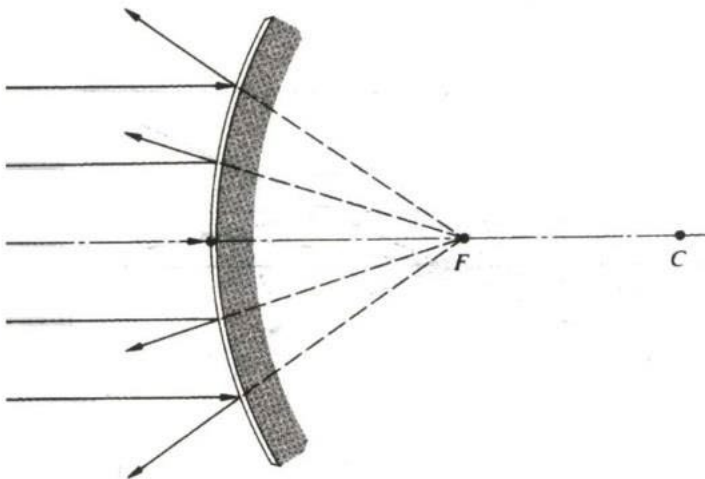
Primary or object focus

$$\lim_{s_i \rightarrow \infty} s_o = f_o = -\frac{R}{2}$$

Secondary or image focus

$$\lim_{s_o \rightarrow \infty} s_i = f_i = -\frac{R}{2}$$

Focus will be positive for concave mirrors ($R < 0$) and negative for convex mirrors ($R > 0$)



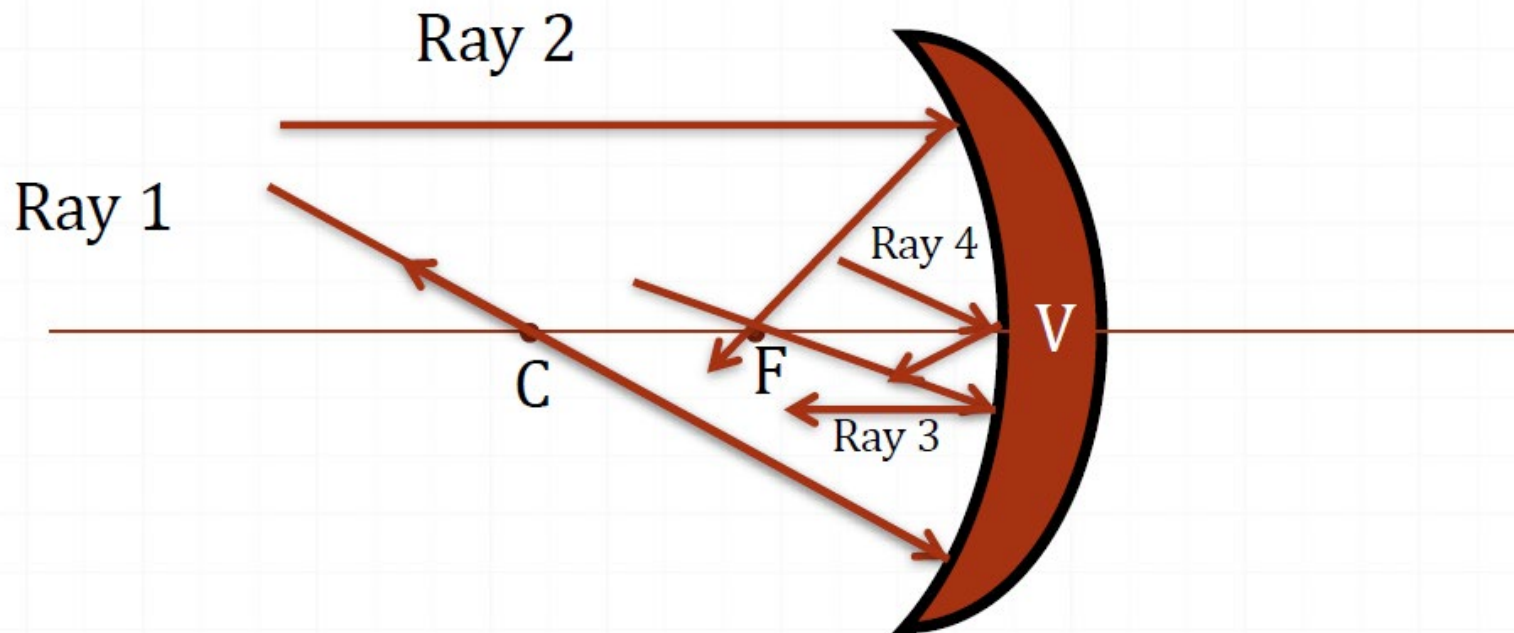
Finite Imagery

Ray 1 passes through the center

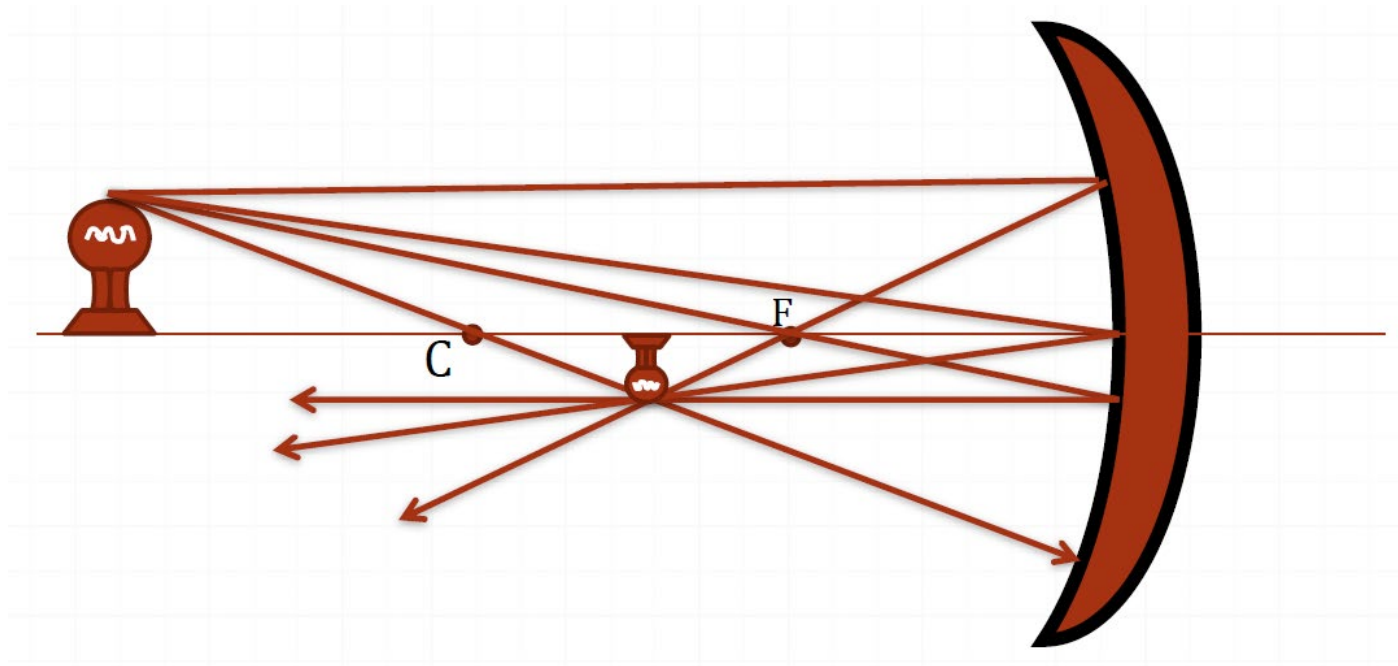
Ray 2 is parallel to the axis

Ray 3 passes through the focus

Ray 4 passes through the vertex

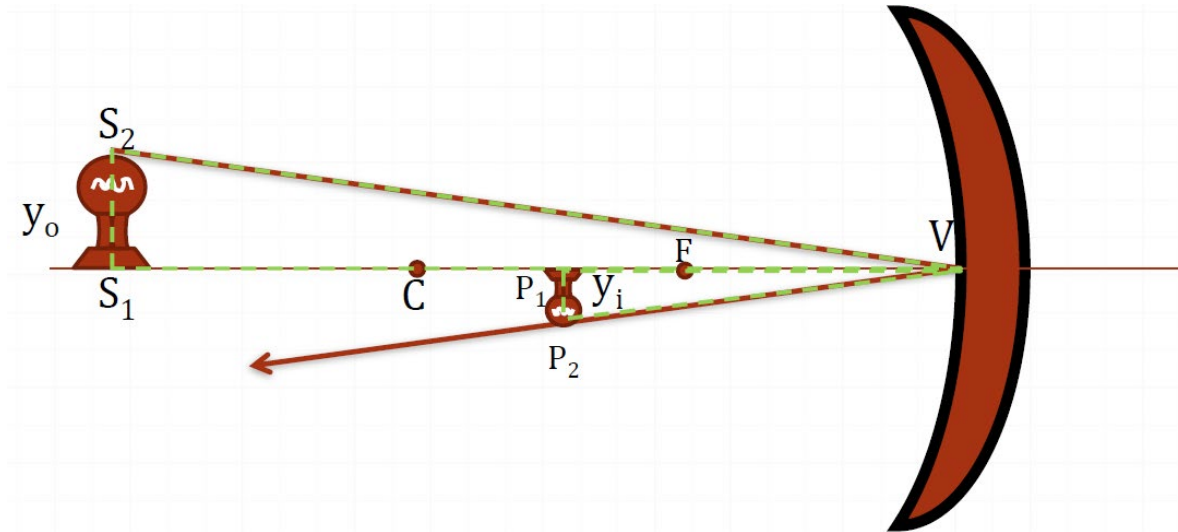


Within the restrictions of paraxial theory, any parallel off-axis bundle of rays will be focused to a point on the focal plane passing through F normal to the optical axis.



Transverse magnification

$$M_T \equiv \frac{y_i}{y_o} = -\frac{s_i}{s_o}$$



The only equation that contains information about the structure of the optical element is that for f , it differs for the thin lens and spherical mirror.

The other functional expressions that relate s_o , s_i , and f or y_o , y_i , and M_T are precisely the same.

The only alteration in the previous sign convention is that s_i on the left of V is now taken as positive.

The striking similarity between

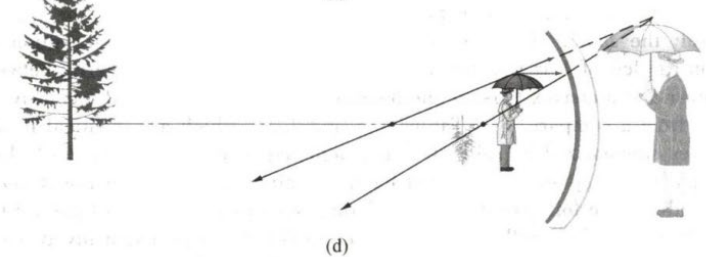
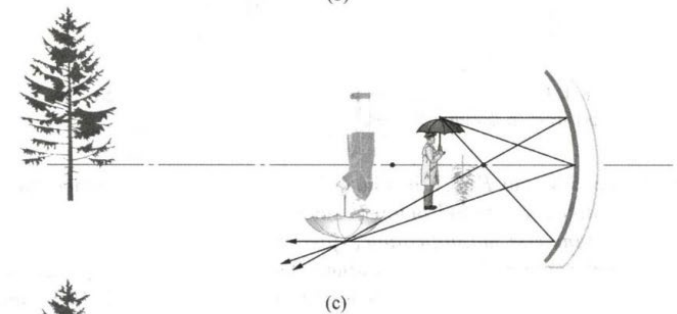
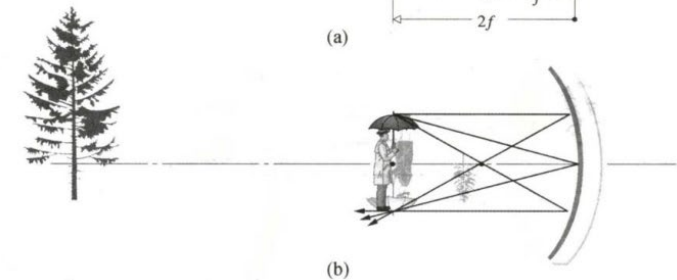
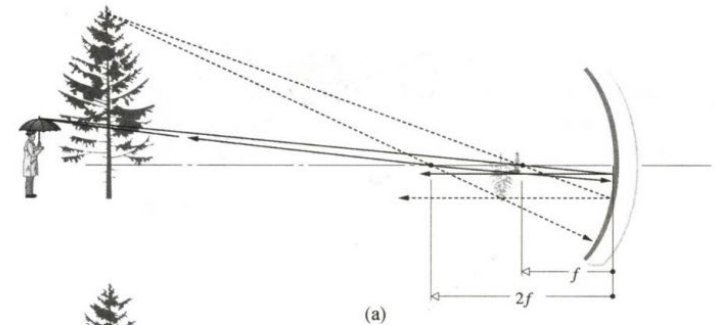
A concave mirror and a convex lens

A convex mirror and a concave lens

$$\frac{1}{s_o} + \frac{1}{s_i} = -\frac{2}{R}$$

Table 4.5 Images of Real Objects Formed by Spherical Mirrors

Concave				
Object		Image		
Location	Type	Location	Orientation	Relative Size
$\infty > s_o > 2f$	Real	$f < s_i < 2f$	Inverted	Minified
$s_o = 2f$	Real	$s_i = 2f$	Inverted	Same size
$f < s_o < 2f$	Real	$\infty > s_i > 2f$	Inverted	Magnified
$s_o = f$		$\pm \infty$		
$s_o < f$	Virtual	$ s_i > s_o$	Erect	Magnified
Convex				
Object		Image		
Location	Type	Location	Orientation	Relative Size
Anywhere	Virtual	$ s_i < f $, $s_o > s_i $	Erect	Minified

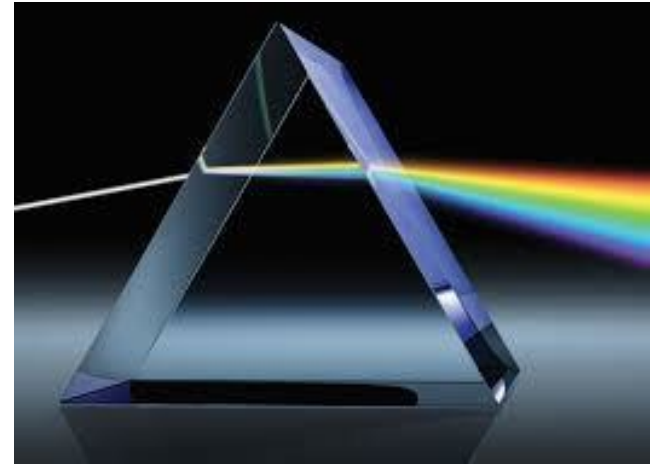


Homework

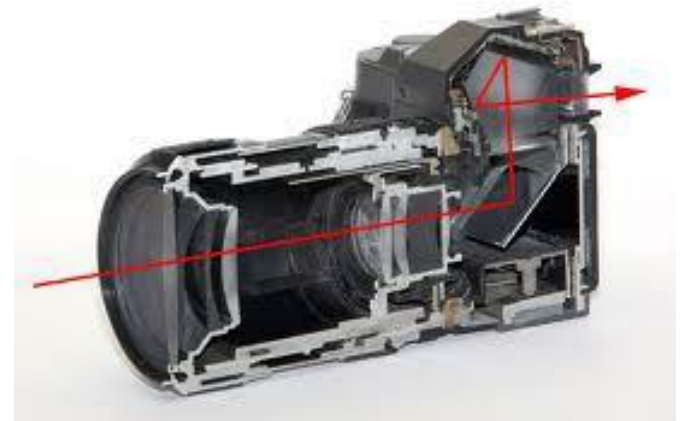
□ 5.44, 5.46, 5.47, 5.52, 5.55, 5.60

4.5 Prisms

- ❑ Dispersive prisms
 - ✓ Separate the constituent frequency components in a polychromatic light beam.



- ❑ Reflecting prisms
 - ✓ Change the orientation of an image or direction of propagation of a beam



Dispersing prisms

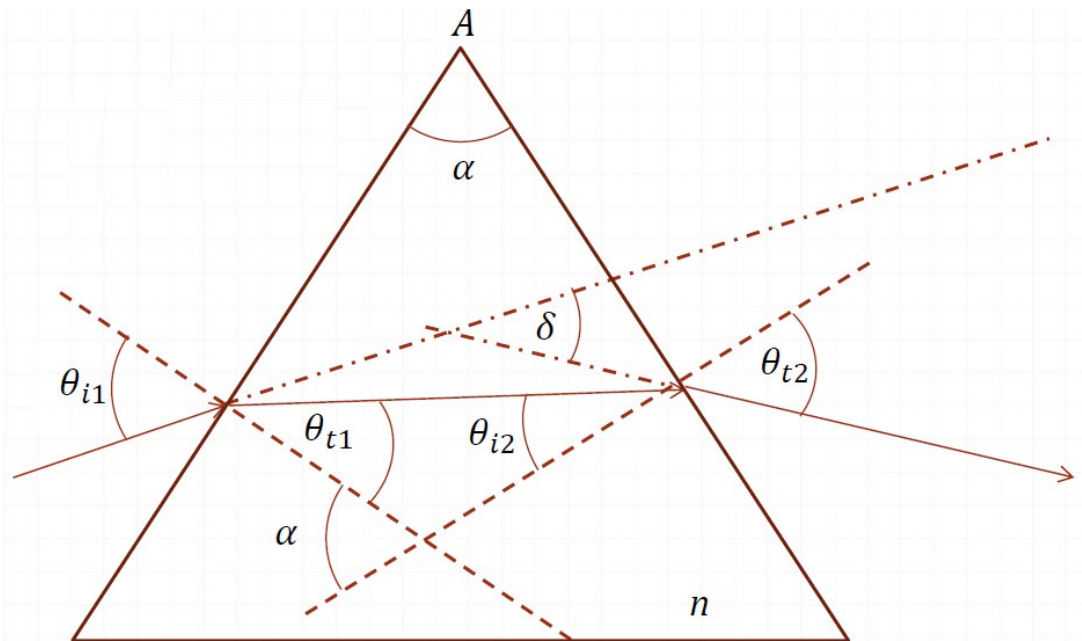
Angular deviation (δ): a ray entering a dispersing prism will emerge having been deflected from its original direction by an angle.

$$\delta = (\theta_{i1} - \theta_{t1}) + (\theta_{t2} - \theta_{i2})$$

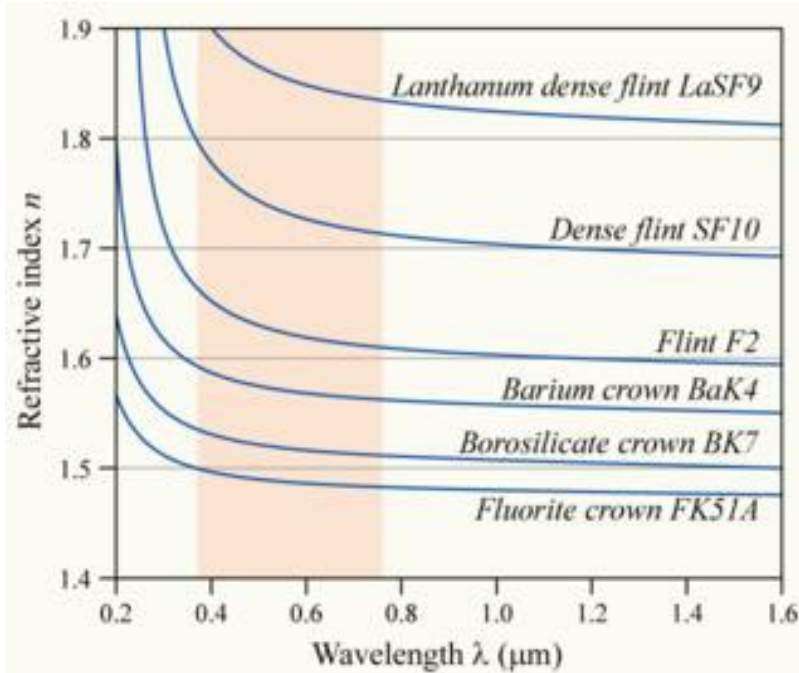
$$\text{Apex angle } \alpha = \theta_{t1} + \theta_{i2} \qquad \delta = \theta_{i1} + \theta_{t2} - \alpha$$

$$\delta = \theta_{i1} + \sin^{-1} \left[\sin \alpha (n^2 - \sin^2 \theta_{i1})^{\frac{1}{2}} - \sin \theta_{i1} \cos \alpha \right] - \alpha$$

δ increases with n , which is itself a function of frequency, so we designate the deviation as $\delta(\nu)$ or $\delta(\lambda)$



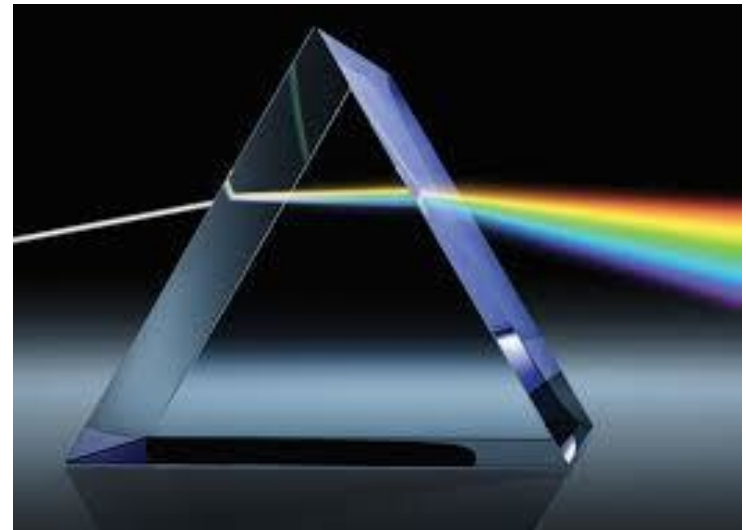
The wavelength dependence of the index of refraction for various materials



$$\delta = \theta_{i1} + \sin^{-1} \left[\sin \alpha (n^2 - \sin^2 \theta_{i1})^{\frac{1}{2}} - \sin \theta_{i1} \cos \alpha \right] - \alpha$$



So $\delta(\lambda)$ will be less for red light than it is for blue



For most transparent dielectric of practical concern, $n(\lambda)$ decreases as the wavelength increases across the visible.